

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, JavaScript, TypeScript, Java, SQL, Bash

Systems & Performance

Linux profiling, Spark / PySpark, Spark UI, performance tuning, multithreading, OpenMP / SIMD, cache-aware design, distributed data processing

Backend & Reliability

Node.js, Express, FastAPI, REST APIs, multi-tenant SaaS, async services, integration testing

ML & Data

PyTorch, JAX, NumPy, deep learning, automatic differentiation, data annotation pipelines, RAG / retrieval

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability for **multi-tenant systems** (Node.js, MongoDB) serving **8+ clients** and processing **10,000+ monthly submissions**; monitored production logs across releases and integrations, investigated incidents, and shipped fixes for tenant-specific issues.
- Architected a full-stack **data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label and manage training data for production models, replacing spreadsheet workflows; cut manual work by **100+ hours/month** and improved labeling throughput by **60%**.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive workflows against regressions, hardening authentication and rate limiting.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability.

Projects

Spark ETL Performance Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI | 1.4B-row NYC TLC dataset

- Cut end-to-end ETL runtime **25%** (17–19 min to 14.3 min) on a **1.4B-row** dataset by replacing shuffle joins with broadcast joins (minutes to 0.05s), enabling AQE and skew-join handling, and tuning shuffle partitions, schema-aware reads, and year/month partitioning.
- Used **Spark UI** stage timings to localize shuffle and skew bottlenecks, cutting tail-latency ratio (P99/P50) from **2.14x to 1.74x**, and documented a repeatable optimization playbook.

Generative Models & Deep Learning - From-Scratch ML Systems

github.com/Deva-1903/UMASS_CICS_689

NumPy, JAX, PyTorch | UMass CS 689 Advanced ML

- Built a matrix-based reverse-mode **automatic differentiation** engine from scratch in NumPy, validated against JAX to machine precision, and trained/benchmarked **5 neural architectures** on CIFAR-10 across 3 optimizers, working with deep-learning systems from first principles.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a cross-platform desktop application integrating camera, microphone, UI, and backend services into an **on-device** pipeline (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS), designed and shipped under hackathon time pressure.
- Engineered a FastAPI backend and Electron/React frontend running fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while streaming local video and audio.

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a Python/FastAPI service that turns open-ended queries into structured tables via typed retrieval planning, page rerank, deterministic extraction with LLM fallback, and output verification, returning cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation for debuggability and reliability.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.